

External IoT Retrofit for Remote Monitoring and Control of a Rheem EW1-50RP Water Heater

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I. INTRODUCTION

DOMESTIC electric storage water heaters may require users to be near the appliance to read its temperature and operate its controls. The Rheem EW1-50RP (50 L, 1500 W) is the reference case for this project. It already measures and displays the temperature at the center of its tank, controls heating through its own thermostat, and offers an AI Smart mode [1]. The missing function addressed here is remote access to the existing panel information and on/off button.

This proposal adds a removable external IoT layer. A camera reads the displayed temperature, an RGB sensor observes the heating pilot, and a flow sensor measures water use at an accessible pipe connection. A servo presses the original on/off button. An application presents the three measurements and sends an on/off request; a simple voice interface uses the same request path. The heater retains its own temperature setting and protections. The prototype is specific to the EW1-50RP, while the underlying access problem can affect other appliances with local-only controls.

II. PROBLEM STATEMENT

A. Problem Description and Justification

For the selected Rheem heater, the user must approach the panel to read the tank-center temperature or press its on/off button [1]. This limits remote preparation of hot water and makes it difficult to check the heater's observed state when away from the appliance. The original proposal also considered electrical and water consumption. The revised prototype retains water-use measurement, but it does not infer electrical energy from a light indicator or a nominal power rating.

Research on domestic water heating shows that scheduling and control can reduce energy use under studied conditions [2]; those results do not establish savings for this heater or this prototype. The immediate, testable problem is remote visibility and control without opening the heater or switching its mains supply. Flow data add a record of when and how much water passes the selected external point, which can help distinguish heating from actual water use [3], [4].

III. THEORETICAL FRAMEWORK

A. Panel Readout and Heater Operation

The EW1-50RP's own controller displays a temperature measured at the center of its tank. The panel includes an on/off button, temperature adjustment buttons, and a heating indicator; red denotes heating and green denotes temperature

preservation [1]. Reading this display avoids estimating the tank temperature from an external pipe, which can be misleading when no water flows.

A camera facing the numeric display can capture the digits without accessing the heater's electronics. Tongbai demonstrated optical reading of a seven-segment instrument using an ESP32-CAM for image capture and server-side recognition [5]. The present design adopts the readout principle, but its accuracy on the Rheem panel must be tested. An RGB light sensor with a small ambient-light shield reads the pilot independently [6].

B. Water Flow and External Actuation

The YF-S201 turbine flow sensor produces Hall-effect pulses as water turns its rotor. Its nominal calibration is approximately 450 pulses per liter; actual volume is obtained by counting pulses and calibrating the installed assembly [7]. The sensor is inserted in a removable external pipe union. This requires work on the plumbing connection, although it does not require opening the heater.

The single actuator is a low-voltage servo that briefly presses and releases the existing momentary on/off button. Mechanical button pushers are a commercial precedent for external appliance control [8]. The servo does not replace the heater's thermostat or thermal protections. Because the on/off button toggles state, a command must be based on a fresh panel observation; an unconfirmed command must not be repeated blindly.

C. Communication and Preprocessing

An ESP32 controller counts flow pulses and classifies the shielded RGB signal. A separate ESP32-CAM captures the visor; numeric recognition runs in the application service, keeping image processing off the control loop. Wi-Fi carries measurements and commands between the nodes and the application. The application accepts explicit on/off requests and simple spoken commands through the same validated route. It displays unavailable or uncertain readings as such instead of guessing a value.

IV. STATE OF THE ART

Eight studies guide the scope of this proposal. Each supports a particular design choice; none alone validates the complete Rheem retrofit.

A1. Smart-home energy conservation

Bhati et al. [9] examine household attitudes and behavior in a smart-city energy context. We take the need for useful feedback and simple interaction. Their study does not measure this heater or demonstrate savings from this prototype.

A2. Control of domestic water heating

Booyesen et al. [2] examine how optimal control can reduce energy for domestic water heating under modeled or studied conditions. This motivates remote access to the heater's on/off function. We do not transfer their savings figures to the EW1-50RP.

A3. Water-use detection without internal access

Singer et al. [10] investigate external flow sensing for a smart water-heater controller. It supports the value of water-use information in this application. Our proposed Hall meter instead requires a removable external plumbing union.

A4. Flow measurements around domestic heaters

Pirow et al. [4] estimate domestic hot-water usage from temperature and vibration sensors outside a pipe. Their proof of concept shows that water-use monitoring can be separated from the heater electronics. It does not validate our Hall sensor or its installation.

A5. Demand and flow data

Lee and Yim [3] analyze domestic hot-water energy and flow demand using smart-meter data. This supports measuring use rather than treating powered time as water consumption. Their apartment-complex results are not a prediction for one household.

A6. Optical reading of a display

Tongbai [5] uses an ESP32-CAM to capture a digital instrument display without a communication port and recognizes the digits on a web server. We apply the same capture-and-recognize approach to the Rheem visor, subject to testing its geometry and lighting.

A7. Advanced scheduling as a boundary

Shen et al. [11] study data-driven electric water-heater scheduling and control. Their work explains why demand information may be useful later. Predictive scheduling is not part of this first prototype.

A8. Usability of smart-home systems

Marikeyan et al. [12] review smart-home literature from the user's perspective. This supports a legible status interface and a small voice-command vocabulary. It does not prove that a voice assistant improves energy efficiency.

A. Proposed Contribution

These sources separately address hot-water control, flow data, optical readout, and user interaction. Our contribution is to combine three external observations with a removable button pusher on the specific EW1-50RP, while keeping the heater's original temperature and electrical controls.

V. OBJECTIVES

General Objective

Develop and evaluate a removable external IoT prototype for remote monitoring and on/off control of a Rheem EW1-50RP water heater, while leaving its internal controller and mains connection untouched.

Specific Objectives

- SO1. Read the temperature shown on the heater's original display with a camera and classify the heating pilot with an RGB sensor.
- SO2. Measure water flow and accumulated volume at an accessible external pipe connection with a Hall-effect flow sensor.
- SO3. Preprocess the three sensor outputs and present temperature, observed heating state, and water use in a wireless application.
- SO4. Press the heater's original on/off button with one removable servo mechanism after a validated user request, then check the observed result.
- SO5. Provide a basic voice interface that converts on/off requests and status questions into the same application actions.
- SO6. Compare the reported data and button response with direct observation during prototype tests.

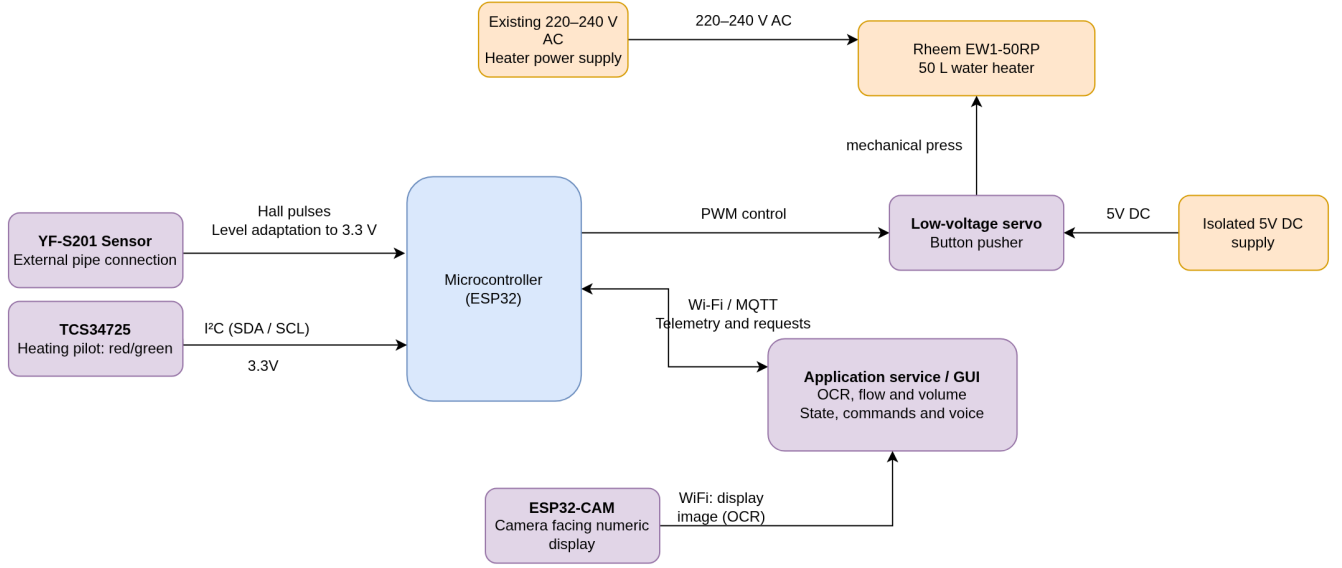
Scope and Limits

The implementation targets one EW1-50RP. Its temperature setpoint is configured on the original panel; remote setpoint adjustment is outside this prototype. The camera reads the value displayed by the heater, not an independent measurement of tank or outlet temperature. If the visor is dark, a current temperature cannot be reported until a safe way to wake it is verified. The color sensor reports the visible pilot state, not electrical current or energy consumption. The flow meter reports water passing its installation point, not total household use. The prototype does not claim energy savings or calculate kWh. The servo uses low-voltage power and does not switch the 220–240 V supply. Plumbing compatibility, installation, and final sensor calibration are part of validation.

VI. METHODOLOGY

A. Reference Appliance and Components

The reference is the Rheem EW1-50RP, a 50 L, 1500 W, 220–240 V storage heater. Its panel already shows tank-center temperature and controls heating [1]. The proposed add-on uses exactly three sensors and one actuator, listed in Table I. The ESP32-WROOM controller and ESP32-CAM camera node are development boards, not additional sensors or actuators.



Removable panel support: camera, color sensor and servo aligned; manual access preserved.
Separate low-voltage supply for the add-on. Flow meter requires an external plumbing union.

Figure 1: External sensing and button actuation. The Rheem 220–240 V supply remains under its original control.

Table I: Proposed external prototype components.

Role	Part	Location and output
Sensor 1: temperature readout	ESP32-CAM / OV2640	Camera facing the numeric visor; captures images for digit recognition.
Sensor 2: pilot state	TCS34725 color sensor (I ² C)	Shielded optical view of the pilot; classifies red, green, or unreadable light.
Sensor 3: water use	YF-S201 Hall flow meter (5 V, G1/2 nominal)	Pulse output for flow rate and volume at a compatible external pipe union [7]; level shifting protects the ESP32's 3.3 V input.
Actuator: on/off	Low-voltage servo with external pusher	One brief press of the original on/off button followed by retraction; torque and stroke chosen after a button test.
Controller and link	ESP32-WROOM-32; Wi-Fi/MQTT	Counts pulses, classifies light, validates commands, and drives the servo.
Interface and enclosure	Application; removable low-voltage case	Displays preprocessed readings; shields optics and houses the electronics without blocking the original panel.

B. Architecture and Operating Sequence

Figure 1 separates sensing and low-voltage actuation from the heater's existing mains supply. The camera node transfers a cropped image of the visor to the application service for digit recognition. The controller counts Hall pulses, derives flow and accumulated volume, and classifies color readings under a light shield. MQTT over Wi-Fi carries telemetry and on/off requests between the ESP32 controller and application service; the camera sends images over Wi-Fi for recognition. The application shows the latest valid temperature, pilot state, and water use, marking stale or unreadable data instead of treating it as current.

For a requested state change, the controller first checks a recent pilot observation. It issues one short servo press only when the observed state differs from the requested state, then retracts and waits for a new observation. A dark or ambiguous pilot does not by itself prove that the heater is off. If the state cannot be confirmed, the application reports uncertainty

and asks for manual verification; it does not keep toggling. Reading the numeric visor may require waking the panel, but the button's wake-versus-toggle behavior is not documented and will be tested before any automated wake press is enabled. A local manual press remains possible after removing the pusher, and loss of Wi-Fi does not alter the heater's original operation.

C. Calibration and Validation

The proposed enclosure uses a detachable support to keep the camera, color sensor, and servo aligned with the panel. Commercial button pushers such as SwitchBot Bot inform the external press-and-release mechanism [8]; their compatibility is not assumed for this heater. The support must allow removal for manual access. Its attachment, dimensions, and servo travel will be selected from the panel geometry and button tests. The detachable panel module is distinct from the flow meter, whose installation requires opening and resealing an

external plumbing union. The proposed YF-S201 has nominal male G1/2 connections [7]; the size and thread type of the heater's flexible connection have not been measured. Before procurement, mating fittings, available space, pressure and temperature ratings, and a location clear of the relief-valve discharge must be verified. An adapter or a different meter may be required.

The camera is aligned and shielded against glare. Recognized digits are compared with the visible panel over several temperatures; blank images and heater error codes are marked unavailable. The color classifier is checked under red, green, dark, and ambient-light conditions. The flow meter is calibrated with a measured water volume; pulses per liter and the installation point are recorded. Finally, the servo stroke and return are tested on the momentary button, and on/off requests are compared with the observed panel response. The electronics use a separate low-voltage supply inside a removable case. The case design leaves the heater's ventilation, safety discharge, and ordinary access unobstructed. The external water union must be checked for fitting size, rated pressure and temperature, leaks, and its position relative to the relief valve before installation.

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